



COURSE SLIDES

Microsoft Azure Security Engineer Associate (AZ-500)

TGS-2023039344 | Hands-On Labs | Tertiary Infotech Academy Pte Ltd | UEN: 201200696W

Version 3

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About Tertiary Infotech Academy

WSQ course provider delivering ICT and business technology training.

UEN: 201200696W

Training format: instructor-led, hands-on, assessment-ready.

About the Trainer



Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation, software engineering and cloud learning.

DELIVERS

WSQ courses on cloud fundamentals, AI, automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Digital Attendance

- Complete SSG TRAQOM digital attendance at the required checkpoints.
- Use the same registered learner identity throughout the course.
- Assessment digital attendance is separate from class attendance.
- The TRAQOM QR code and LMS submission are provided through the LMS/TMS portal.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

Learner Introduction

- Share your name, role and Azure security experience.
- State one identity, network or monitoring control you want to practise.
- Confirm your access to the assigned Azure subscription or sandbox.

Course Objectives

Identity

Labs 1-2

RBAC, PIM, Conditional
Access and managed
identities

Network

Labs 3-4

NSG, ASG, Firewall, WAF and
Private Endpoint controls

Platform

Labs 5-6

Bastion, JIT, encryption, SQL,
Storage and Key Vault

Operations

Labs 7-8

Defender for Cloud, Policy,
Sentinel and playbooks

Briefing for Assessment

- Assessment is open book and conducted through the LMS.
- Complete WA first, then PP.
- Use lab evidence and screenshots where instructed.
- Ask for clarification before the assessment starts, not during submission.

Assessment & Funding

Written Assessment

1 hr

Open book. Complete the WA before PP.

Practical Performance

1 hr

Use the LMS and lab evidence as instructed.

Digital Attendance

Required

Take assessment attendance before starting.

Funding

SSG

Meet attendance and assessment requirements.

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Access the Hands-On Labs

<https://github.com/tertiarycourses/TGS-2023039344---Microsoft-Azure-Security-Engineer-Associate-AZ-500-/tree/main/labs>

Option A

Clone the repository

```
git clone  
https://github.com/tertiarycourses/TGS-  
2023039344---Microsoft-Azure-Security-  
Engineer-Associate-AZ-500-.git
```

Option B

Download the ZIP

GitHub Code menu > Download ZIP, then open the labs folder

Run

Use the browser

Follow each lab in Azure Portal using the instructor-provided subscription

Record screenshots or notes where the lab asks for evidence. If a feature is unavailable because of licensing or permission limits, document the exact blocker.

Courseware & Assessment on the LMS

- Download the courseware and submit assessment answers at <https://lms-tms.tertiaryinfotech.com/>.
- Use the LMS for TRAQOM QR code, assessment attendance and submission instructions.
- Keep your learner account active until all assessment records are signed.

Two-Day Lesson Flow



Day 1: Labs 1-4 - identity, privileged access, Conditional Access, managed identities and network protection.

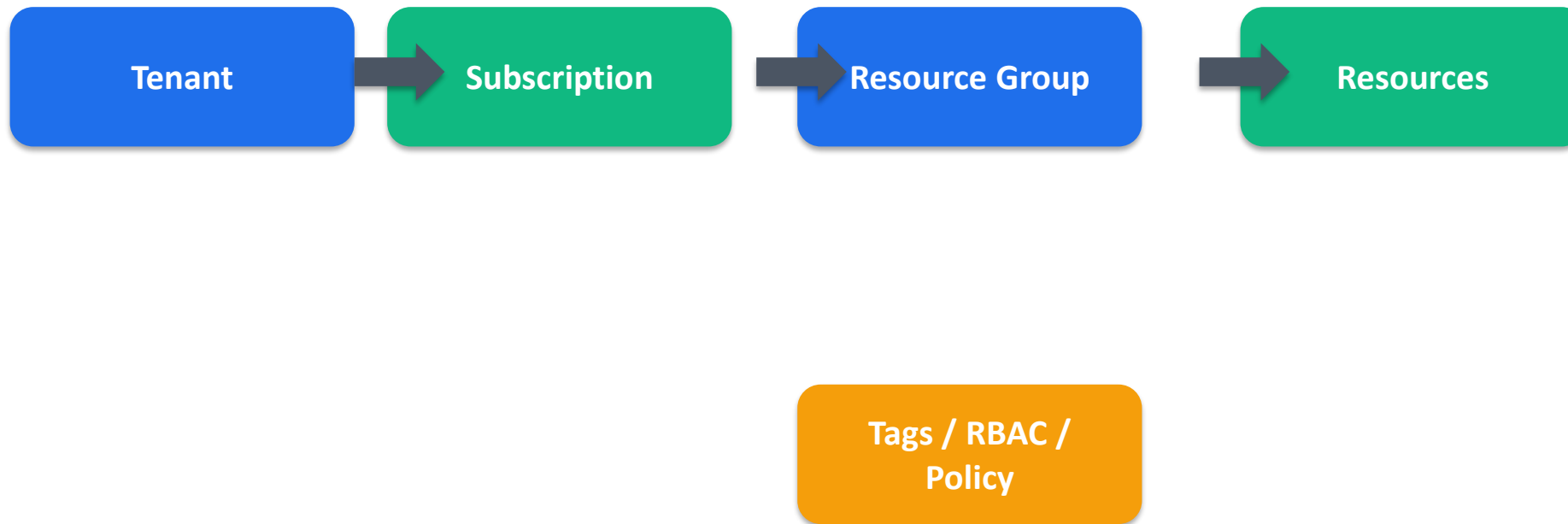


Day 2: Labs 5-8 - compute, data protection, Defender for Cloud, Sentinel investigation and automation.



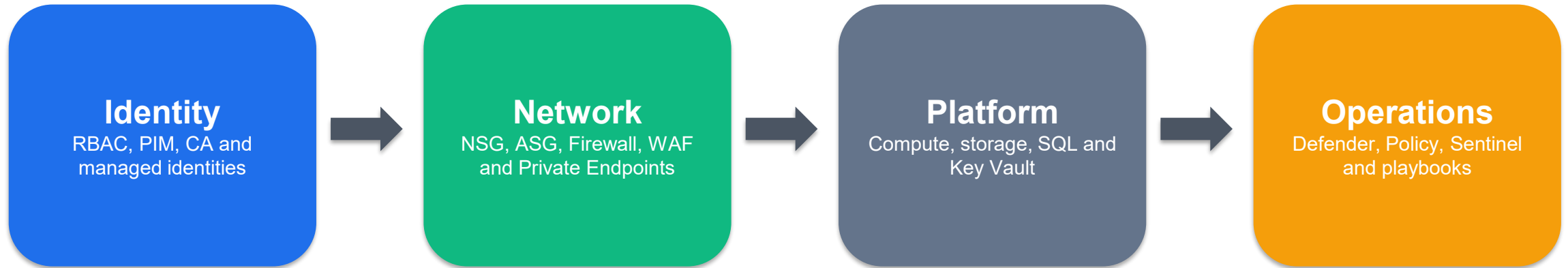
All lab work uses the assigned training subscription and `rg-az500-<initials>` unless your trainer instructs otherwise.

Azure Resource Model



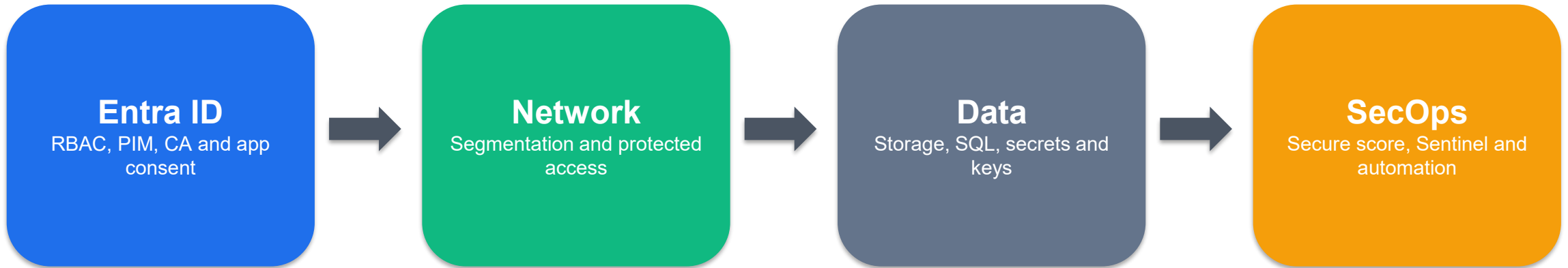
The labs repeatedly use this hierarchy: sign in to the tenant, work inside a subscription, place resources in rg-az500-
<initials>, then apply identity, network, workload, data and SecOps controls.

Two-Day Azure Lab Flow



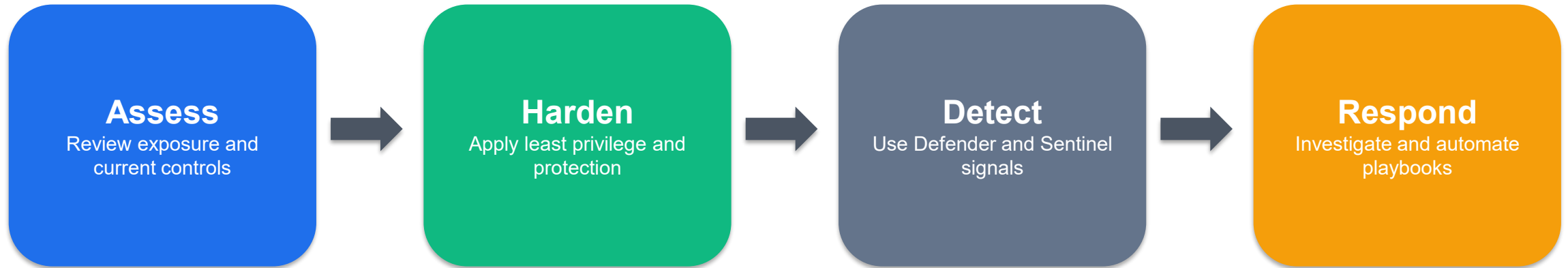
The slide deck, Learner Guide and Lesson Plan follow this same lab sequence.

Azure Services Practised



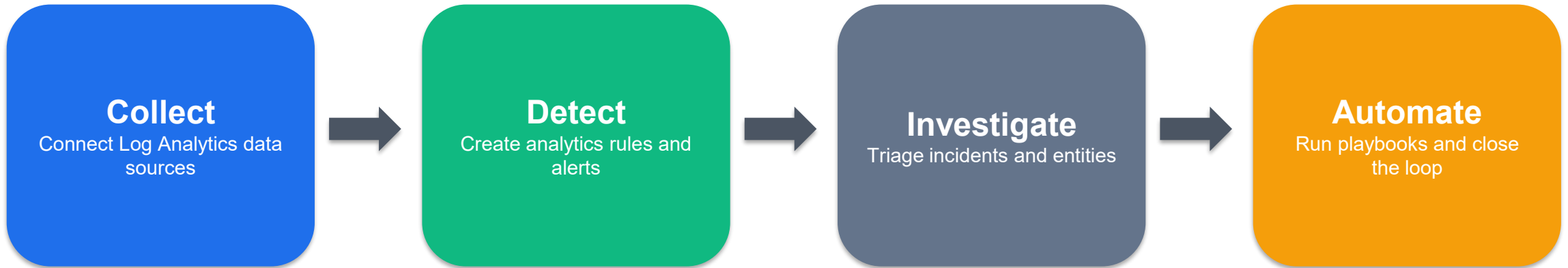
Service groups in the diagram map directly to Labs 1-8 and the AZ-500 lab domains.

Security Posture Management Loop



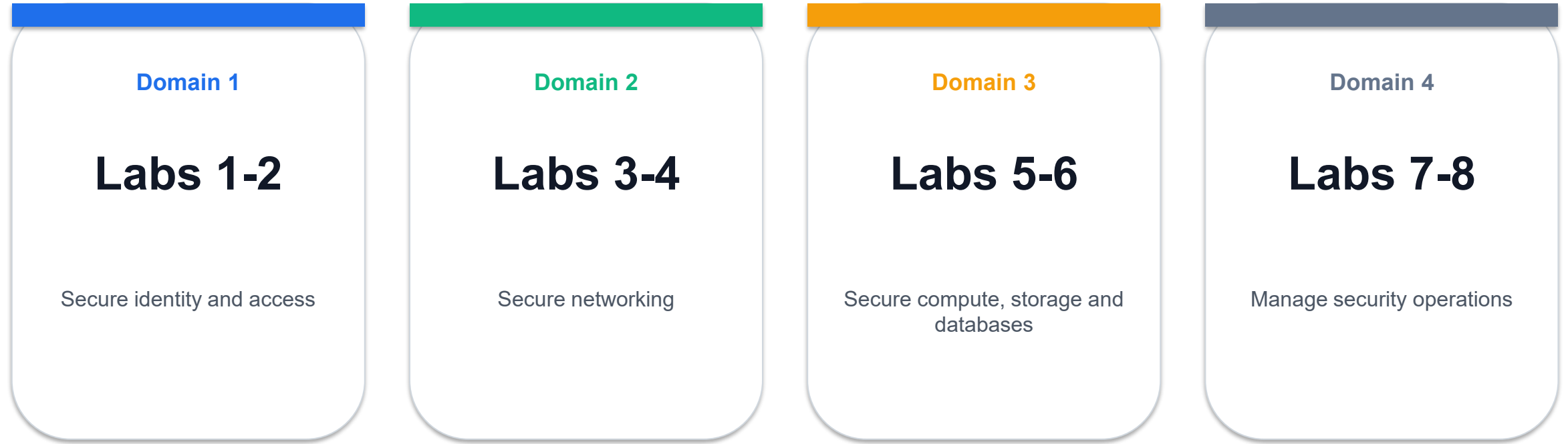
The course moves from prevention to detection, investigation and response.

Sentinel Investigation Lifecycle



Lab 8 closes the course with monitoring, investigation and automation.

AZ-500 Lab Domain Map



AZ-500 Lab Domain Matrix

Domain	Labs	Security Outcome	Priority
Identity and access	1-2	Least privilege and secure app access	High
Networking	3-4	Segmentation and protected ingress	High
Platform and data	5-6	Compute, storage, SQL and secrets	High
Security operations	7-8	Posture, detection and response	High

This matrix keeps the deck, Learner Guide and Lesson Plan aligned to the eight AZ-500 labs.

Azure Defense-in-Depth Layers



Learners move from preventive controls to detection, investigation, automation and evidence capture.

Lab 1 - Configure RBAC, Custom Roles, and Privileged Access Review

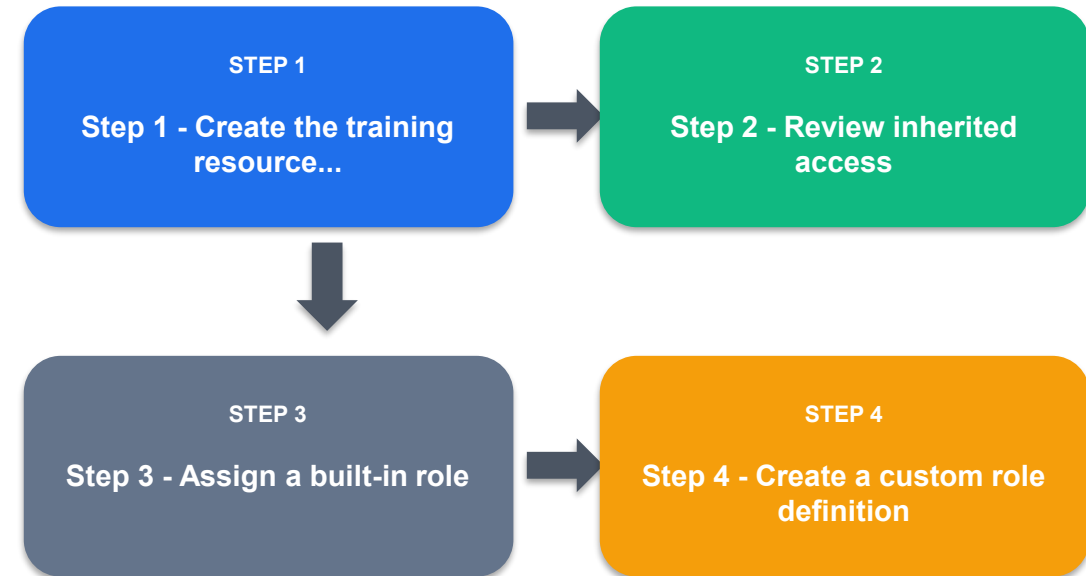
Domain 1 - Secure identity and access

In this lab you will configure Azure role-based access control, create a scoped custom role, and review privileged access patterns with Microsoft Entra Privileged Identity Management.

Day 1

[labs/lab-01-identity-rbac-pim.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 1 - Key Steps



Step 1 - Create the training resource group



Step 2 - Review inherited access



Step 3 - Assign a built-in role



Step 4 - Create a custom role definition



Step 5 - Review privileged identity management

Lab 1 - Verify and Clean Up

Verification

1. How Azure RBAC scopes permissions.
2. How built-in and custom roles differ.
3. How PIM reduces standing privilege.
4. How to document missing identity licensing or permissions during security reviews.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 2 - Secure App Access with Conditional Access and Managed Identities

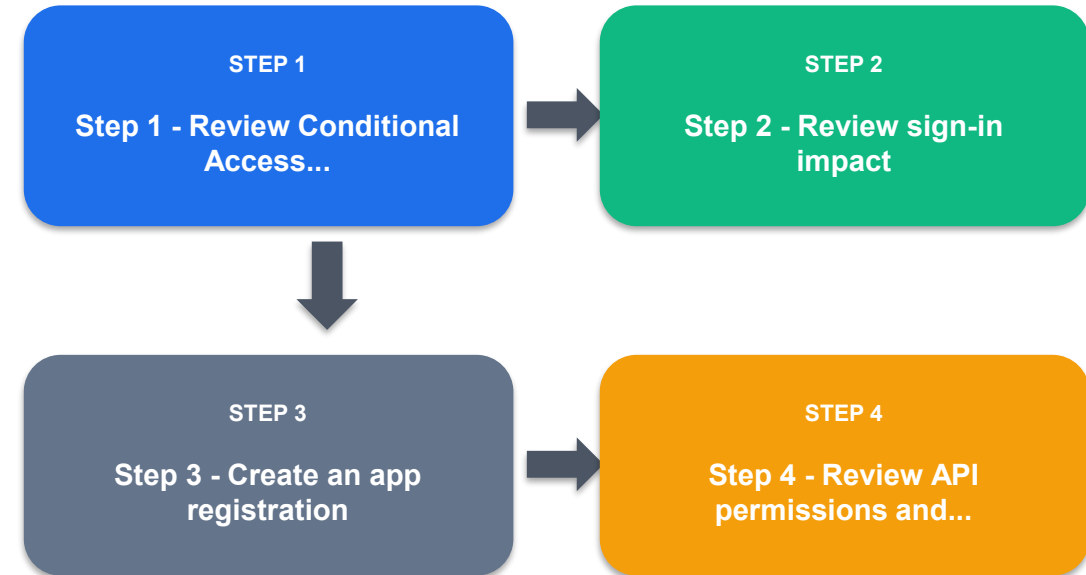
Domain 1 - Secure identity and access

In this lab you will review Conditional Access, app registrations, service principals, OAuth consent, and managed identities. These controls are central to secure identity and application access in Azure.

Day 1

[labs/lab-02-conditional-access-managed-identities.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 2 - Key Steps



Step 1 - Review Conditional Access policies



Step 2 - Review sign-in impact



Step 3 - Create an app registration



Step 4 - Review API permissions and consent



Step 5 - Enable a managed identity

Lab 2 - Verify and Clean Up

Verification

1. How report-only Conditional Access protects against accidental lockout.
2. How app registrations expose API permissions and consent decisions.
3. How managed identities replace application secrets.
4. How service principals connect applications to Azure RBAC.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 3 - Secure Virtual Networks with NSGs, ASGs, Peering, and Network Watcher

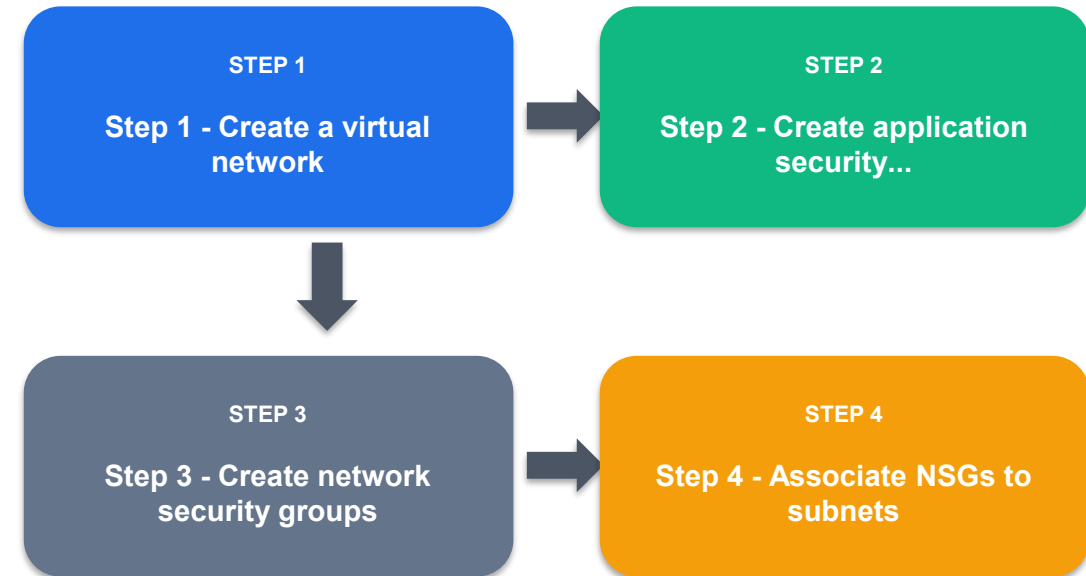
Domain 2 - Secure networking

In this lab you will create a segmented virtual network, apply Network Security Groups, use Application Security Groups, and validate traffic using Network Watcher.

Day 1

[labs/lab-03-network-security-nsg-asg.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 3 - Key Steps



Step 1 - Create a virtual network



Step 2 - Create application security groups



Step 3 - Create network security groups



Step 4 - Associate NSGs to subnets



Step 5 - Use Network Watcher IP flow verify

Lab 3 - Verify and Clean Up

Verification

1. How NSGs enforce subnet and NIC-level traffic rules.
2. How ASGs make rules easier to manage.
3. How Network Watcher validates network security behavior.
4. Why least privilege network design uses explicit tier flows.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 4 - Protect Public and Private Access with Azure Firewall, WAF, and Private Endpoints

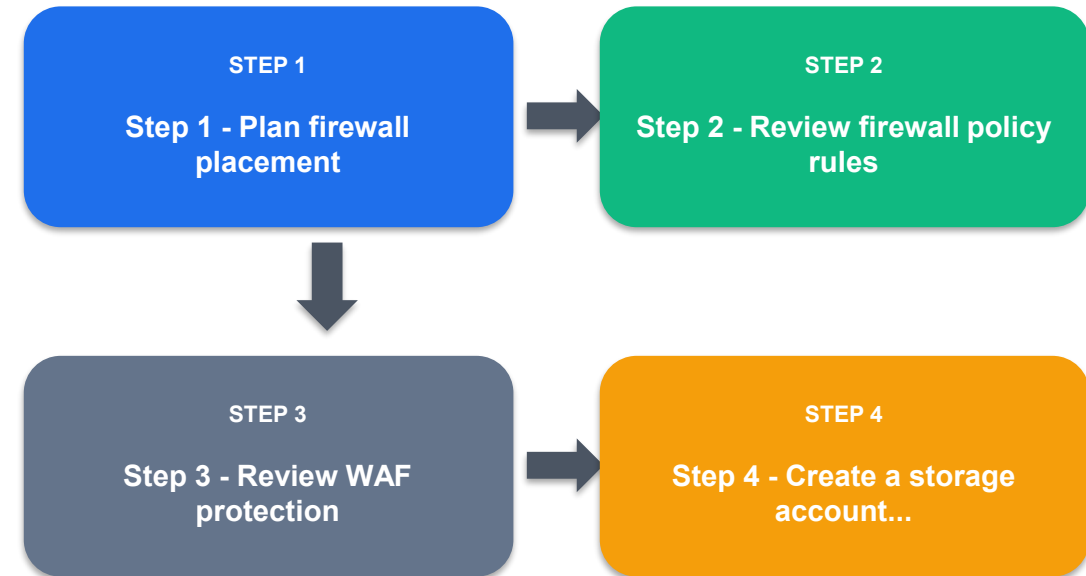
Domain 2 - Secure networking

In this lab you will design public and private access protections using Azure Firewall, Web Application Firewall concepts, and Private Endpoints. Some resources can be reviewed instead of deployed if...

Day 1

labs/lab-04-firewall-
waf-private-
endpoints.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 4 - Key Steps



Step 1 - Plan firewall placement



Step 2 - Review firewall policy rules



Step 3 - Review WAF protection



Step 4 - Create a storage account private endpoint



Step 5 - Review public access settings

Lab 4 - Verify and Clean Up

Verification

1. When to use Azure Firewall, WAF, and Private Endpoints.
2. How firewall policies organize network and application rules.
3. How private access reduces public attack surface.
4. How to document cost-sensitive security resources without deploying every component.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 5 - Secure Virtual Machines, Bastion, JIT Access, and Disk Encryption

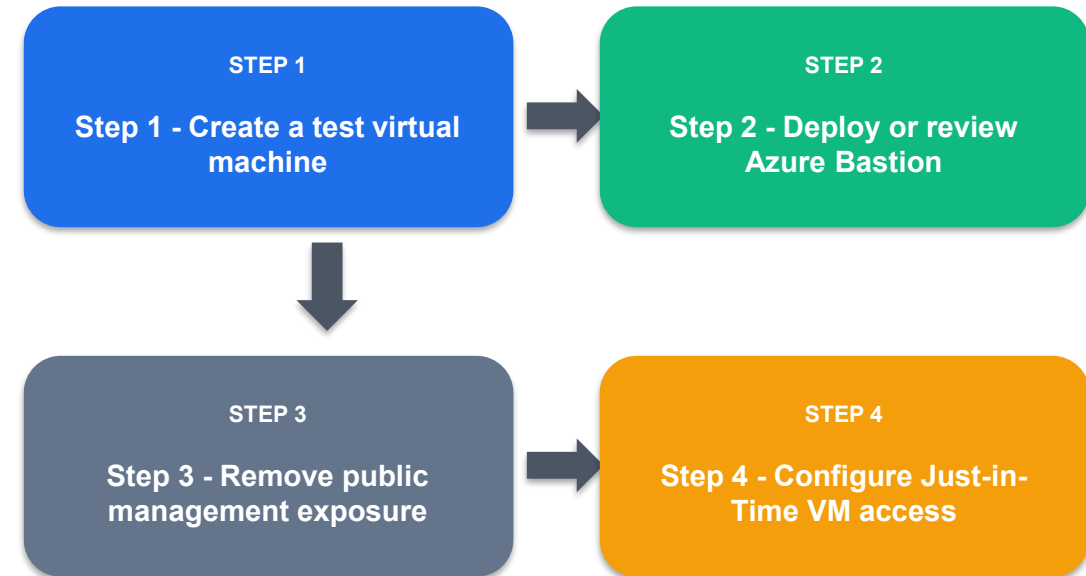
Domain 3 - Secure compute, storage, and databases

In this lab you will secure administrative access to virtual machines and review compute hardening options such as Bastion, Just-in-Time access, and disk encryption.

Day 2

labs/lab-05-
compute-security-
vm-bastion-jit.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 5 - Key Steps



Step 1 - Create a test virtual machine



Step 2 - Deploy or review Azure Bastion



Step 3 - Remove public management exposure



Step 4 - Configure Just-in-Time VM access



Step 5 - Review disk encryption options

Lab 5 - Verify and Clean Up

Verification

1. How Bastion reduces VM management exposure.
2. How JIT limits management ports to approved time windows.
3. How NSGs, Defender for Cloud, and encryption combine for VM security.
4. How to choose between platform-managed and customer-managed encryption.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 6 - Protect Storage, SQL Database, and Key Vault Secrets

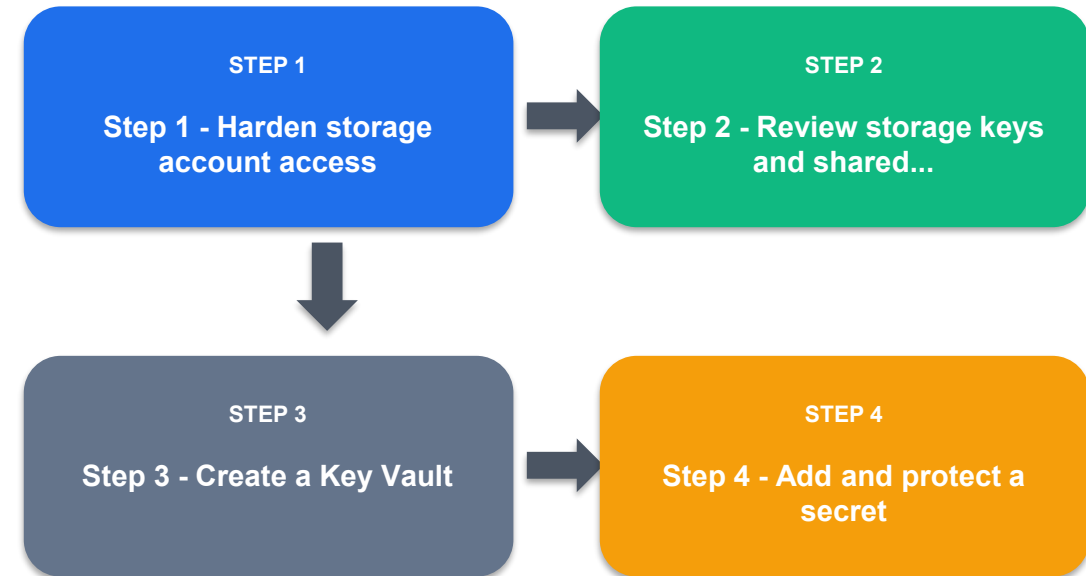
Domain 3 - Secure compute, storage, and databases

In this lab you will configure security controls for Azure Storage, Azure SQL Database, and Azure Key Vault.

Day 2

[labs/lab-06-storage-sql-key-vault.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 6 - Key Steps



Step 1 - Harden storage account access



Step 2 - Review storage keys and shared access



Step 3 - Create a Key Vault



Step 4 - Add and protect a secret



Step 5 - Create or review Azure SQL Database security

Lab 6 - Verify and Clean Up

Verification

1. How to harden storage access and recovery settings.
2. How Key Vault protects keys, secrets, and certificates.
3. How Azure SQL Database security features map to confidentiality and audit requirements.
4. Why secret values should not be copied into lab submissions.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 7 - Manage Security Posture with Azure Policy and Defender for Cloud

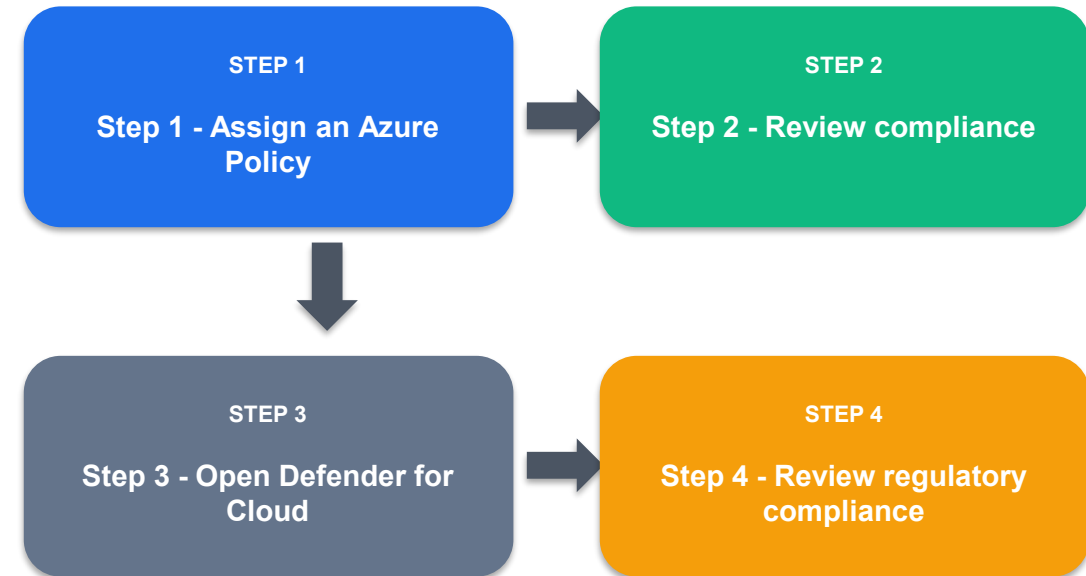
Domain 4 - Manage security operations

In this lab you will use Azure Policy and Microsoft Defender for Cloud to assess compliance, secure score, and recommended remediation actions.

Day 2

labs/lab-07-
defender-policy-
secure-score.md

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 7 - Key Steps



Step 1 - Assign an Azure Policy



Step 2 - Review compliance



Step 3 - Open Defender for Cloud



Step 4 - Review regulatory compliance



Step 5 - Review Defender plans

Lab 7 - Verify and Clean Up

Verification

1. How Azure Policy enforces governance.
2. How Defender for Cloud calculates secure score.
3. How recommendations support remediation.
4. How compliance standards map technical controls to frameworks.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Lab 8 - Monitor, Investigate, and Automate with Microsoft Sentinel

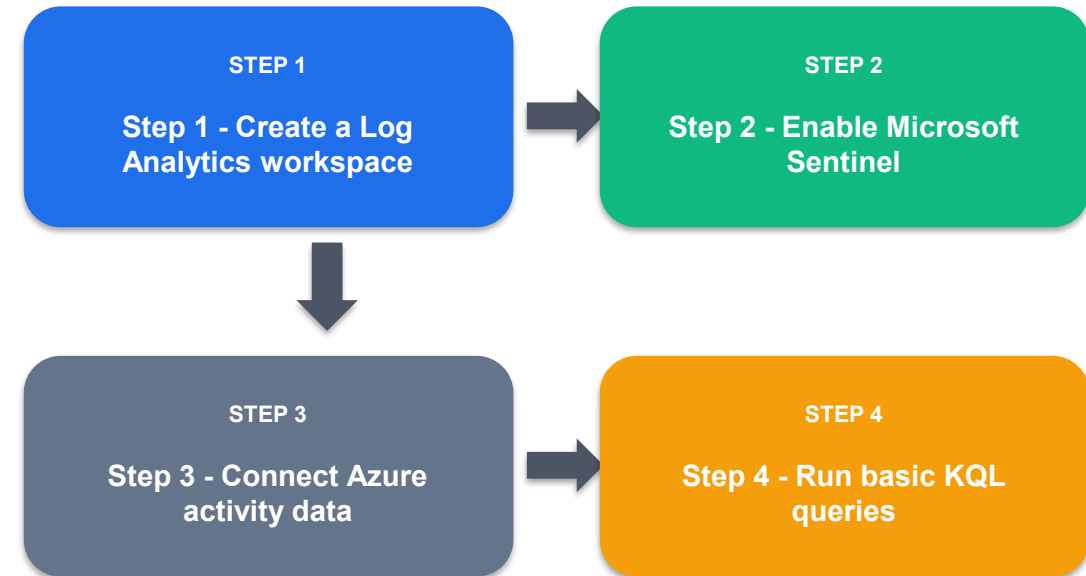
Domain 4 - Manage security operations

In this lab you will configure a Microsoft Sentinel workspace, connect data, run KQL queries, create an analytics rule, and review automation options.

Day 2

[labs/lab-08-sentinel-monitor-investigate-automate.md](#)

Hands-on



Follow the Learner Guide exactly for commands, portal fields, verification and cleanup.

Lab 8 - Key Steps



Step 1 - Create a Log Analytics workspace



Step 2 - Enable Microsoft Sentinel



Step 3 - Connect Azure activity data



Step 4 - Run basic KQL queries



Step 5 - Create an analytics rule



Step 6 - Review automation

Lab 8 - Verify and Clean Up

Verification

1. How Sentinel uses Log Analytics workspaces.
2. How data connectors bring telemetry into the SIEM.
3. How KQL supports investigation.
4. How analytics rules and automation support alerting and response.

Cleanup Discipline

1. Run the lab cleanup section exactly.
2. Keep `rg-az500-<initials>` only when the next lab needs it.
3. Confirm no paid resource is left running.
4. Lab 8 and the trainer's final instructions cover course cleanup.

Course Wrap-Up

-  You configured identity, network, compute, data and security operations controls across the eight labs.
-  You practised evidence-based security review when licensing or permission limits blocked deployment.
-  Delete remaining training resources unless the instructor asks you to preserve them for assessment.

Take the Online AZ-500 Practice Exams

- Reinforce AZ-500 domains with realistic, exam-style questions before assessment day.
- Use practice sets to test identity, network, platform, data and security operations controls.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to review.

Access and practise at:

[Google Classroom > Practice Exams](#)

The screenshot displays the 'Tertiary Exams' website interface. At the top, there is a navigation bar with a blue circular logo, the text 'Tertiary Exams', and links for 'Practice Exam' and 'About Us'. A blue 'Get started' button is located in the top right corner. The main heading is 'Microsoft Azure Security Engineer Associate (AZ-500) Practice Exams'. Below this, there are three interactive cards: 'MODE' with a blue bar and the text 'Practice', 'FOCUS' with a green bar and the text 'AZ-500', and 'ACCESS' with a purple bar and the text 'Class link'. To the right of these cards is a 'PRACTICE DETAILS' box containing the text: 'Exam code AZ-500', 'Use Review', and 'Domains 4'.

Assessment

Written Assessment

1 hr

Open book. Complete the WA before PP.

Practical Performance

1 hr

Use the LMS and lab evidence as instructed.

Digital Attendance

Required

Take assessment attendance before starting.

Funding

SSG

Meet attendance and assessment requirements.

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Digital Attendance (Mandatory)

- Complete SSG TRAQOM digital attendance at the required checkpoints.
- Use the same registered learner identity throughout the course.
- Assessment digital attendance is separate from class attendance.
- The TRAQOM QR code and LMS submission are provided through the LMS/TMS portal.

Thank You

-  Submit all assessment responses on the LMS.
-  Sign the Assessment Summary Record.
-  Contact Tertiary Infotech Academy for post-course support.